

Question 1 (15 Marks) Begin a SEPARATE sheet of paper

Marks

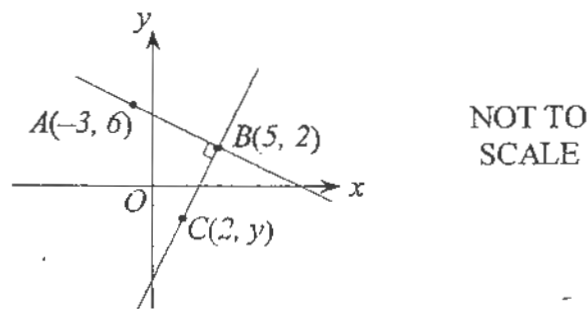
(a) Graph on the number line the solution set of: $|2x - 1| \geq 7$. 2

(b) Simplify:

(i) $\frac{3\log_e 8}{\log_e 16}$ 1

(ii) $e^{2\log_e 2}$ 1

(c)



The diagram shows the origin O and the points $A(-3, 6)$, $B(5, 2)$ and $C(2, y)$. The lines AB and BC are perpendicular. A and B lie on the line $x + 2y = 9$ and the length of AB is $4\sqrt{5}$ units.

(i) Find the perpendicular distance from O to AB . 2

(ii) Show that C has the coordinates $(2, -4)$. 1

(iii) Does the line AC pass through the origin? Explain. 2

(d) Sketch the curve $y = 1 + \sin x$ for $0 \leq x \leq 2\pi$ 1

(e) Shade the region in the Cartesian number plane which represent the simultaneous solution of $y \leq \sqrt{4 - x^2}$ and $y \geq |x| - 2$. 2

(f) Find the locus of a point $P(x, y)$ which moves in Cartesian number plane such that it is equidistant from points $A(-2, -4)$ and $B(6, 2)$. 3

Question 2 (15 Marks) Begin a SEPARATE sheet of paper

Marks

- (a) Simplify fully $\frac{2}{x^2 - 9} - \frac{1}{x - 3} - \frac{4}{x + 3}$. 2
- (b) For what values of r will the quadratic equation $x^2 + (r - 2)x + 1 = 0$ has unreal roots? 3
- (c) Find the derivative of the following:
- (i) $\log_e(6x^2 - 3)$. 1
- (ii) $\sin^3 x$. 1
- (iii) $x^2 \ln x$. 2
- (d) Given that $y = \frac{e^{2x}}{2x + 1}$, show that $\frac{dy}{dx} = \frac{4xe^{2x}}{(2x + 1)^2}$. 2
- (e) If $f(x) = \frac{1}{\sqrt{2x + 7}}$, find $f'(x)$. 2
- (f) Let α and β be the roots of the equation $x^2 - rx + s = 0$, where r and s are real numbers and not equal to zero. Find
- (i) $\alpha + \beta$ 1
- (ii) $\frac{1}{\alpha} + \frac{1}{\beta}$ 1

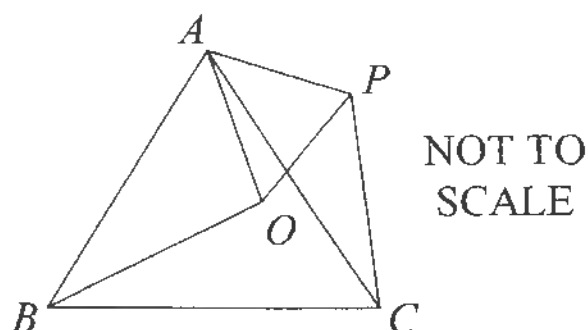
Question 3 (15 Marks) Begin a SEPARATE sheet of paper

Marks

(a) Prove that $\frac{\cot \theta \operatorname{cosec} \theta}{1 + \cot^2 \theta} = \cos \theta$. 2

(b) Solve for x $9^x - 10(3^x) + 9 = 0$. 2

(c) In the figure shown triangles ACB and APC are equilateral.



(i) Copy this diagram onto your answer sheet and explain why $\angle BAO = \angle PAC$. 1

(ii) Prove $\triangle AOB \equiv \triangle APC$. 2

(d) The function $y = x^3 - 6x^2 + 9x - 4$ is defined in the domain $0 \leq x \leq 4$.

(i) Find the co-ordinates of any stationary points and determine their nature. 3

(ii) Draw a neat sketch of the function and indicate the appropriate position of the point of inflexion. 3

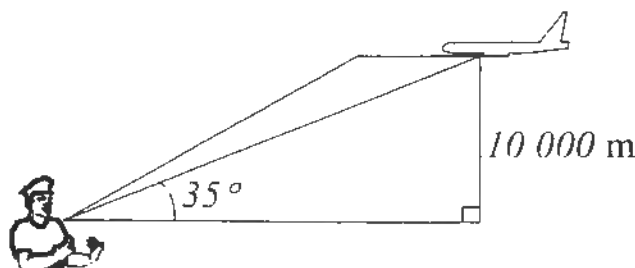
(e) Show that $\frac{(a+b-2\sqrt{ab})}{\sqrt{a}-\sqrt{b}}$ is a rational number, given that a and b are two distinct positive rational numbers. 2

Question 4 (15 Marks) Begin a SEPARATE sheet of paper

Marks

- (a) A plane is flying horizontally at a height of 10 000 m above a level stretch of ground at a constant speed of 240 km/h. A man on the ground observes the angle of elevation of the plane (which is flying on a course directly towards him) to be 35° . How far (to the nearest km) in a straight line is the man from the plane 1 minute after he measured the angle of elevation?

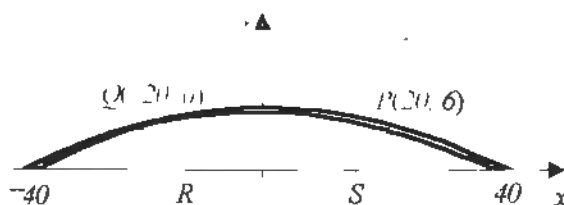
3



- (b) A heavy metal arch is to be fixed in the vertical plane as shown in the figure below.

The arch is in the shape of a parabola with equation $y = -\frac{x^2}{200} + 8$

Two straight beams are to be fixed at the points P and Q to hold the arch in position. For maximum support the beams must be normal to the parabolic arch.



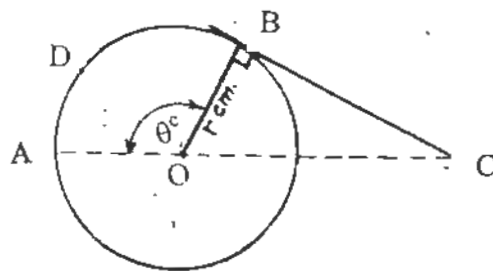
- (i) Find the position of the point R where the beam will be fixed at ground level on the x axis.
- (ii) Find (to the nearest degree) the angle of inclination of the beam QR .

3

2

Question 4 continued

- (c) In the figure shown, arc ADB plus the interval BC equals x cm
(ie. arc ADB + interval $BC = x$ cm). The radius of the circle is r cm.



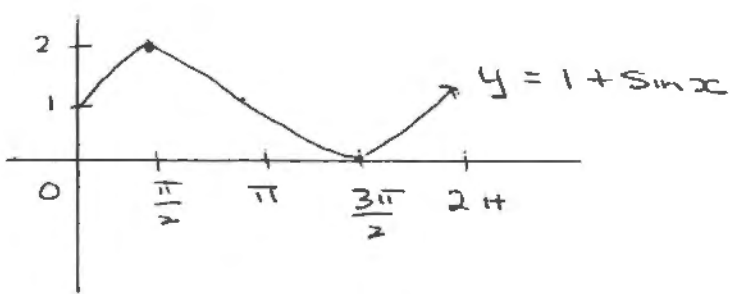
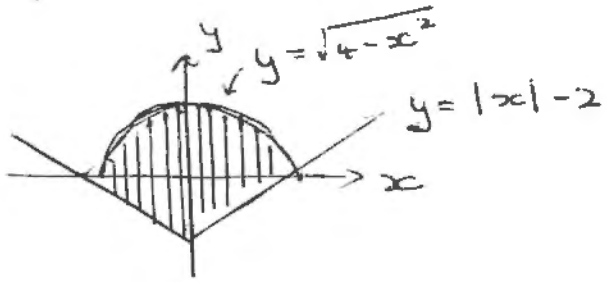
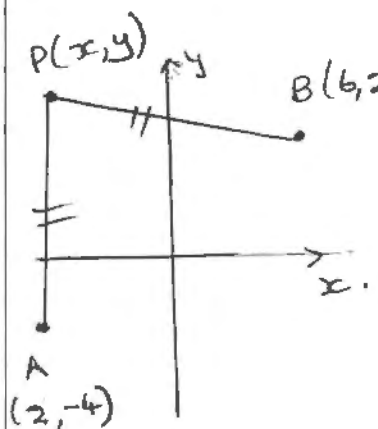
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|-------|--|---|
| (i) | If A, O, C are collinear points and angle $AOB = \theta^\circ$,
show that $\theta - \tan \theta = \frac{x}{r}$ | 3 |
| (ii) | If the area of sector AOB is $\frac{3}{8}$ of the area of the circle, find θ and the ratio $\frac{x}{r}$. | 2 |
| (iii) | Express AC in terms of r and θ . | 2 |

End of paper

MATHEMATICS: Question ... 1 ..

Suggested Solutions	Marks	Marker's Comments
a) $2x+1 \geq 7$ $2x+1 \geq 7$ or $-(2x+1) \geq 7$ $x \geq 3$ or $x \leq -4$ 	1 1	
b (i) $\frac{3 \log 8}{\log 16} = \frac{a \log 2}{4 \log 2} = \frac{9}{4}$	1	
(ii) $e^{2 \log 2} = 4$	1	
c (i) $d = \frac{ ax_1 + by_1 + c }{\sqrt{a^2 + b^2}}$ $a=1$ $b=2$ $c=-9$ $x_1=0$ $y_1=0$ $d = \frac{ 1(0) + 2(0) - 9 }{\sqrt{1^2 + 2^2}}$ $= \frac{ -9 }{\sqrt{5}}$ $= \frac{9}{\sqrt{5}} \text{ units or } \frac{9\sqrt{5}}{5} \text{ units}$	1 1	
ii) $M_{AB} = \frac{2-6}{5+3} = \frac{-4}{8} = -\frac{1}{2}$ $\therefore M_{CB} = 2$ $\frac{y_2 - y_1}{x_2 - x_1} = 2$ $= \frac{y-2}{2-5} = 2$ $= \frac{y-2}{-3} = 2$ $= y-2 = -6$ $= y = -4$ $\therefore \text{point C is } (2, -4)$	1	

MATHEMATICS: Question 1 continued.

Suggested Solutions	Marks	Marker's Comments
c (iii) $MAC = \frac{-4-6}{2+3} = \frac{-10}{5} = -2$ Equation $y - y_1 = m(x - x_1)$ $y - 6 = -2(x + 3)$ $y = -2x$ Yes $(0,0)$ lies on the line	1 1	
d) 	1	
e) 	1 1	graph shading.
f)  $PA = PB$ $\sqrt{(x+2)^2 + (y+4)^2} = \sqrt{(x-6)^2 + (y-2)^2}$ $(x+2)^2 + (y+4)^2 = (x-6)^2 + (y-2)^2$ $x^2 + 4x + 4 + y^2 + 8y + 16 = x^2 - 12x + 36 + y^2 - 4y + 4$ $4x + 4 + 8y + 16 = -12x + 36 - 4y + 4$ $16x + 12y - 20 = 0$ $\text{or } 4x + 3y - 5 = 0$ $\text{or } y = \frac{-4x + 5}{3}$	1 1 1	

MATHEMATICS: Question...2..

Suggested Solutions	Marks	Marker's Comments
a) $\frac{2}{x^2-9} - \frac{1}{x-3} - \frac{4}{x+3}$ $= \frac{2 - (x+3) - 4(x-3)}{x^2-9}$ $= \frac{-5x+11}{x^2-9} \quad \text{or} \quad \frac{-5x+11}{(x+3)(x-3)}$	1 1	
b) $\Delta = b^2 - 4ac < 0$ $= (r-2)^2 - 4(1)(1) < 0$ $(r-2)^2 < 4$ $r-2 < \pm 2$ $r-2 < 2 \quad \text{or} \quad r-2 > -2$ $r < 4 \quad \text{or} \quad r > 0$ 	1 1 1	$a=1$ $b=r-2$ $c=1$
c) (i) $\log_e (6x^2-3)$ $\frac{f'(x)}{f(x)} = \frac{12x}{6x^2-3} \quad \text{or} \quad \frac{12x}{3(2x^2-1)} \quad \text{or} \quad \frac{4x}{2x^2-1}$	1	
(ii) $\sin^3 x = (\sin x)^3$ $\frac{dy}{dx} = 3(\sin x)^2 \cdot \cos x$ $= 3 \cos x (\sin x)^2$ $= 3 \cos x \cdot \sin^2 x$	1	
iii) $x^2 \ln x$ let $u = x^2$ so $u' = 2x$ $v = \ln x$ so $v' = \frac{1}{x}$ Product rule $uv' + vu'$ $= 2x \ln x + x^2 \cdot \frac{1}{x}$ $= 2x \ln x + x$	1 1	

MATHEMATICS: Question...2... continued.

Suggested Solutions	Marks	Marker's Comments
d) $y = \frac{e^{2x}}{2x+1}$ quotient rule $u = e^{2x}$ so $u' = 2e^{2x}$ $v = 2x+1$ so $v' = 2$ $\frac{vu' - uv'}{v^2} = \frac{(2x+1) \cdot 2e^{2x} - 2e^{2x}}{(2x+1)^2}$ $= \frac{2e^{2x}[(2x+1)-1]}{(2x+1)^2}$ $= \frac{2e^{2x}(2x)}{(2x+1)^2}$ $= \frac{4xe^{2x}}{(2x+1)^2}$	1	
e) $\frac{1}{\sqrt{2x+7}} = (2x+7)^{-\frac{1}{2}}$ $\frac{dy}{dx} = -\frac{1}{2}(2x+7)^{-\frac{3}{2}} \cdot 2$ $= \frac{-1}{\sqrt{(2x+7)^3}}$	1	
f) i) $\alpha + \beta = \frac{-b}{a} = \frac{-(-r)}{1} = r$ ii) $\frac{1}{\alpha} + \frac{1}{\beta} = \frac{\alpha + \beta}{\alpha\beta} = \frac{r}{s}$	1	

MATHEMATICS: Question 3...

Suggested Solutions

Marks

Marker's Comments

a) $\frac{\cot \theta \operatorname{cosec} \theta}{1 + \cot^2 \theta} = \cos \theta$

LHS

$$\frac{\frac{\cos \theta}{\sin \theta} \times \frac{1}{\sin \theta}}{\cos^2 \theta + \sin^2 \theta}$$

$$= \frac{\cos \theta}{\sin^2 \theta} \div \cos^2 \theta + \sin^2 \theta$$

$$= \frac{\cos \theta}{\sin^2 \theta} \times \sin^2 \theta$$

$$= \cos \theta$$

$$= \text{RHS}$$

$\left(\frac{1}{2}\right) \cot \theta = \frac{\cos \theta}{\sin \theta}$

$\left(\frac{1}{2}\right) \operatorname{cosec} \theta = \frac{1}{\sin \theta}$

$\left(\frac{1}{2}\right) 1 + \cot^2 \theta = \operatorname{cosec}^2 \theta$

$\left(\frac{1}{2}\right) \text{ Prove LHS = RHS}$

b) $9^x - 10(3^x) + 9 = 0$

$(3^x)^2 - 10(3^x) + 9 = 0$

Let $u = 3^x$

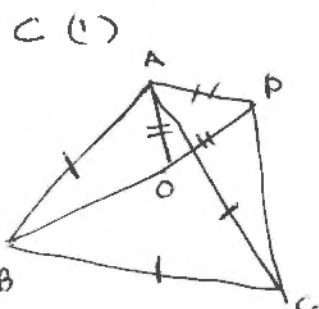
$u^2 - 10u + 9 = 0$

$(u-9)(u-1) = 0$

$u = 9 \text{ or } u = 1$

$3^x = 9 \text{ or } 3^x = 1$

$x = 2 \text{ or } x = 0$



$\hat{BAO} = 60 - \angle AOC$

$\hat{PAC} = 60 - \angle AOC$

Hence both angles are equal to $60 - \hat{AOC}$

ii) $AB = AC$ (given data)
 $AO = AP$ (given data)
 $\angle BAO = \angle PAC$ (from part (i))
 $\therefore \triangle AOB \cong \triangle APC$ (SAS)

MATHEMATICS: Question 3 continued

Suggested Solutions

Marks

Marker's Comments

d) i) $y = x^3 - 6x^2 + 9x - 4$

At TP, $y' = 0$

$$y' = 3x^2 - 12x + 9 = 0$$

$$= 3(x^2 - 4x + 3) = 0$$

$$3(x-3)(x-1) = 0$$

$$x = 3 \text{ or } x = 1$$

when $x=1$ $y = 1 - 6 + 9 - 4 = 0$ so TP(1,0)

when $x=3$ $y = 27 - 54 + 27 - 4 = -4$ so TP(3,-4)

Nature $y'' = 6x - 12$

at $x=1$ $y'' = -ve$ so, $\nearrow$ Max TP

x	0	1	2
$\frac{dy}{dx}$	+	0	-

at $x=3$ $y'' = +ve$ so $\searrow$ Min. TP

x	2	3	4
$\frac{d^2y}{dx^2}$	-	0	+

ii) Sketch

domain when $x=0, y=4$
 $x=4, y=0$

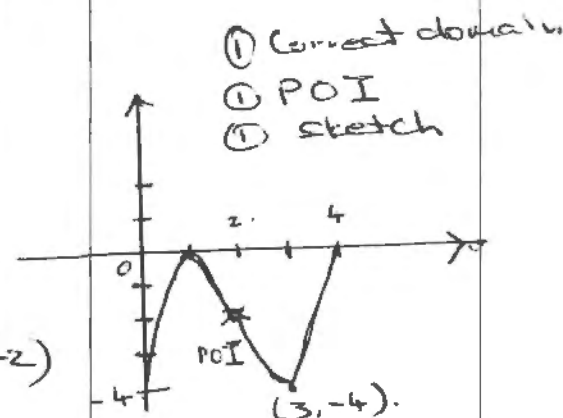
point of inflexion occurs $y'' = 0$

$$6x = 12$$

$$x = 2$$

when $x=2 \Rightarrow y = 8 - 24 + 18 - 4 = -2$

$\therefore$ pt of inflexion occurs at (2, -2)



e) $(a+b-2\sqrt{ab})$

$$\sqrt{a} - \sqrt{b}$$

$$= \frac{(\sqrt{a} - \sqrt{b})^2 \times \sqrt{a} + \sqrt{b}}{\sqrt{a} - \sqrt{b} \quad \sqrt{a} + \sqrt{b}}$$

$$= \frac{(\sqrt{a} - \sqrt{b})(a-b)}{a-b}$$

$$= \sqrt{a} - \sqrt{b}$$

1) recognise

$$a+b-2\sqrt{ab} = (\sqrt{a} - \sqrt{b})^2$$

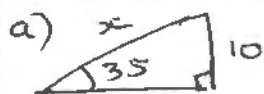
1)

MATHEMATICS: Question 4

Suggested Solutions

Marks

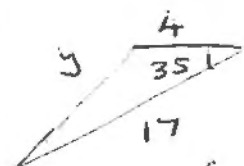
Marker's Comments



$$\sin 35 = \frac{10}{x}$$

$$x = 17.43446796$$

$$x = 17 \text{ km.}$$



Note 1 hr = 240 km
 $\therefore 1 \text{ min} = 4 \text{ km}$

Cosine Rule

$$y^2 = 4^2 + (17.43446796)^2 - 2 \times 4 \times (17.43446796) \cos 35$$

$$= 205.7055324$$

$$y = 14.3425532$$

$$y = 14 \text{ km (nearest km).}$$

b)

$$y = \frac{-x^2 + 8}{200}$$

$$y' = \frac{-2x}{200} = \frac{-x}{100}$$

at point R (-20, 6)

$$y' = \frac{20}{100} = \frac{1}{5} \quad \text{so } m = \frac{1}{5}$$

$$\therefore m_{\text{normal}} = m_{\text{OR}} = -5.$$

Equation of QR

$$y - y_1 = m(x - x_1)$$

$$y - 6 = -5(x + 20)$$

$$y = -5x - 94$$

Point R is the x-intercept so put $y = 0$

$$0 = -5x - 94$$

$$5x = -94$$

$$x = -18.8$$

$$\therefore \text{pt R is } (-18.8, 0)$$

ii)

$$\tan \theta = -5$$

$$\theta = -79^\circ$$

with positive x axis $\theta = 180 - 79$
 $= 101^\circ$

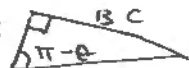
MATHEMATICS: Question 4 continued.

Suggested Solutions

Marks

Marker's Comments

C (i) $x = \text{arc } ADB + BC.$

$= r\theta + r$ 

$\tan(\pi - \theta) = \frac{BC}{r}$

so $BC = r \tan(\pi - \theta).$

$\therefore x = r\theta + r \tan(\pi - \theta)$

$x = r(\theta + \tan(\pi - \theta))$

$= r(\theta - \tan \theta)$

so $\theta - \tan \theta = \frac{x}{r}$

ii) Area

$\frac{1}{2} r^2 \theta = \frac{3\pi r^2}{8}$

$\frac{1}{2} \theta = \frac{3\pi}{8}$

$\theta = \frac{3\pi}{4}$

From (i) $\frac{x}{r} = \theta - \tan \theta$

$\frac{x}{r} = \frac{3\pi}{4} - (-1)$

$\frac{x}{r} = \frac{3\pi + 4}{4}$ or $\frac{3x + 4}{4}$

iii) length $AC = r + OC.$

find length OC

$\cos(\pi - \theta) = \frac{r}{OC}$

$OC = \frac{r}{\cos(\pi - \theta)} = \frac{r}{-\cos \theta}$ or $-r \sec \theta.$



so length $AC = r + (-r \sec \theta)$
 $= r(1 - \sec \theta)$

or
 $r(1 - \frac{1}{\cos \theta})$

$\tan(\pi - \theta) = -\tan \theta$